



AI MEETS POWER BI

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THE SHIFT

AI IS CHANGING HOW WE WORK WITH DATA

*The tools have changed.
The need for judgment hasn't.*

BEFORE

Manual report
building, visual by
visual



NOW

Natural language →
instant report drafts

BEFORE

Digging through
model to understand it



NOW

Ask the assistant to
explain the model

BEFORE

DAX written from
scratch every time



NOW

AI drafts the measure,
you validate it

BUT THERE IS A LOT OF TOOLS...

*The tools have changed.
The need for judgment hasn't.*

BEFORE

Manual report building, visual by visual



NOW

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THE SHIFT

TODAY...



COPILOT

What it is, where it shows up,
licensing & setup



DEMO 1

Copilot in action



MCP SERVERS

Architecture, setup, clients,
tools, resources



DEMO 2 (FOLLOW ALONG)

Install & connect an MCP
server



Fabric Apps

Architecture, setup, clients,
tools, resources



DEMO 3 (FOLLOW ALONG)

Install & connect an MCP
server



PART 1

COPILOT FOR POWER BI

What it is • where it shows up
licensing • prepping data for AI



WHAT IS COPILOT IN POWER BI?



A set of AI-powered experiences built into Power BI that understand your semantic model and help you build, explore, and explain insights in natural language.

Powered by Azure OpenAI
Native in Power BI Service, Desktop & apps
Not one feature — a set of experiences



REPORT SUMMARIES

Explain what a page shows in plain language



DAX SUGGESTIONS

Draft and explain measures on demand



Q&A & NARRATIVE

Ask questions, get smart visual answers



SMART INSIGHTS

Surface anomalies and trends automatically

WHERE COPILOT SHOWS UP



STANDALONE COPILOT

Dedicated conversational workspace — chat with data across your whole org, pick any dataset or report to attach to



COPILOT INSIDE REPORTS IN POWER BI SERVICE

Summaries, visual explanations and guided Q&A for end users already inside the report they're viewing



COPILOT INSIDE AN APP

Summaries, visual explanations and guided Q&A for end users already inside the report they're viewing



POWER BI DESKTOP

Generate and refine DAX, chat with your data while modeling, get report authoring assistance right in Desktop



DEMO 1

LET'S TAKE A LOOK

LIVE WALKTHROUGH – WATCH & TAKE NOTES

KEY TENANT SETTINGS TO KNOW



Users can use Copilot and features powered by Azure OpenAI

The master on/off switch. Enable for whole org or specific security groups. Everything else depends on this.

KEY



Data sent to Azure OpenAI can be processed outside your region

Required when your capacity is outside the EU data boundary or US. Allows cross-region processing.



Capacities can be designated as Fabric Copilot Capacities

Lets capacity admins assign a dedicated Copilot capacity – so users don't need workspace access.



Users can access standalone cross-item Power BI Copilot

Controls the standalone chat experience. Roll out to a test group first before org-wide.



Only show approved items in standalone Copilot

Copilot only answers using data explicitly approved for AI use. Recommended starting point for governance.

KEY

LICENSING & SETUP REQUIREMENTS



WHAT YOU NEED

- F2 capacity or above — OR — P1 / P2 Premium
- Capacity in a supported region (EU / US) — or enable cross-region in tenant settings
- "Users can use Copilot" enabled in Admin Portal
- User has access to the underlying data
- ⚠ NOT available on Fabric trial licenses
- ⚠ NOT supported on sovereign clouds (currently)



FABRIC COPILOT CAPACITY

When users don't have workspace access (e.g. business users consuming apps), assign them to a dedicated Fabric Copilot Capacity instead.

- Set up in Azure portal → Microsoft Fabric
- Start at F2, scale up as needed
- Enable in Admin Portal: "Capacities can be designated as Copilot Capacities"
- Consolidates all Copilot billing into one place
- Monitor via Fabric Capacity Metrics app

PREPPING YOUR DATA FOR AI



Copilot is only as good as the data you give it.



1 SIMPLIFY THE DATA SCHEMA

Prep Data for AI
→ *Simplify Data Schema.*

- Deselect columns Copilot doesn't need
- Remove keys, internal IDs, unused language variants
- Focus on the fields that answer real questions



2 VERIFIED ANSWERS

Right-click any visual
→ *Set Up Verified Answer.*

- Pin specific visuals to common questions
- "Which products sold the most?" always shows the right visual
- Improves consistency across all reports using that model



3 AI INSTRUCTIONS

Prep Data for AI
→ *Add AI Instructions.*

- Tell Copilot what the report is for and who uses it
- Define your org's terminology and metric names
- Tell it what NOT to answer — prevents hallucinated data

WHAT GOOD DATA LOOKS LIKE FOR AI

✓ DO THIS

- Star schema: fact + dimension tables
- Tables named 'Sales', 'Date', 'Customer'
- Field descriptions in the model
- Explicit measures with clear names
- Q&A synonyms configured for key terms
- Internal / technical columns hidden

✗ AVOID THIS

- Flat wide tables with 50+ columns
- Columns named 'Col1', 'Fld_001'
- Undocumented calculated columns
- Implicit measures from raw fields
- Mixed-grain fact tables
- No relationships — just VLOOKUP logic

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PART 2

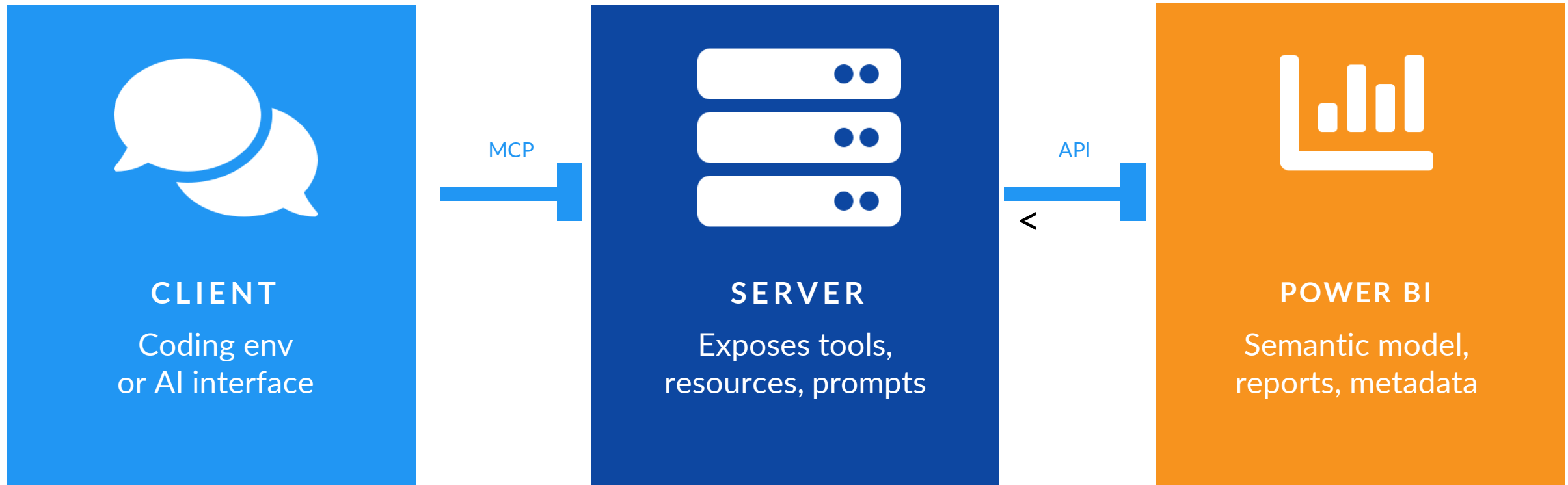
MCP SERVERS FOR POWER BI

What • setup • use cases
best practices • limits



WHAT IS MCP?

Model Context Protocol — a structured way for AI assistants to discover and use tools, context, and resources from external servers.



MCP ARCHITECTURE: KEY CONCEPTS



THE ASSISTANT

Sits between client and server — uses the protocol to reason about which tools to call and when



REQUEST / RESPONSE

Client asks → server replies in structured form → assistant interprets and acts



PROMPTS

Pre-defined prompt templates that guide repeated or complex workflows



AUTHENTICATION

Access follows the authenticated identity — MCP does not bypass permissions or governance

THE MCP SERVERS OF POWER BI

Power BI MCP server (remote)

- Query data and generate insights from existing models

Power BI MCP server (local)

- Build and modify semantic models programmatically

LET'S INSTALL THE REMOTE MCP

1. Download Visual Studio Code
2. <https://learn.microsoft.com/en-us/power-bi/developer/mcp/remote-mcp-server-get-started>

Learn > Microsoft Fabric > Power BI >

Ask Learn

Get started with remote Power BI MCP server (preview)

Summarize this article for me

Important

This feature is in preview.

The remote Power BI MCP server is a remote-hosted endpoint that enables AI agents to chat with data in Power BI semantic models using natural language. Built on the Model Context Protocol (MCP), it translates your prompts into Power BI operations - generating DAX queries and executing them while respecting your permissions and security policies.

Install Power BI MCP server (remote)

LET'S INSTALL THE LOCAL MCP

1. Download Visual Studio Code
2. <https://marketplace.visualstudio.com/items?itemName=analysis-services.powerbi-modeling-mcp>



Power BI Modeling MCP Server

Microsoft  microsoft.com |  158,058 installs |  (10) | Free

The Power BI Modeling MCP Server, brings Power BI semantic modeling capabilities to your agents.

Install

[Trouble Installing?](#) 



DEMO 2

LET'S QUERY OUR SEMANTIC MODEL

LIVE WALKTHROUGH – FOLLOW ALONG IF YOU WANT OR WATCH & TAKE NOTES

WHAT ABOUT SKILLS?

Skills are focused, reusable capabilities that help AI complete specific tasks more effectively.

- They provide task-specific guidance for common workflows.
- They help the AI understand what to do, how to do it, and what to prioritize.
- They can improve consistency across modeling, reporting, design, and management tasks.
- They are especially useful when you want repeatable results instead of generic responses.

POWER BI SKILLS

These skills support the core steps involved in building and managing Power BI content in Fabric.

- **Semantic model authoring:** create, edit, deploy, and optimize Power BI semantic models.
- **Power BI report authoring:** create, edit, and validate reports in PBIR format.
- **Power BI report design:** define a structured design brief for report layout and visual direction.
- **Power BI report planning:** plan and build a new report from an existing semantic model.
- **Power BI report management:** publish, retrieve, and manage reports in Fabric

workspaces.

LET'S INSTALL IT

1. Install Copilot CLI: <https://github.com/features/copilot/cli>

winget install GitHub.Copilot

2. Run Copilot by typing «Copilot»

3. Install the Skills for Fabric: <https://github.com/microsoft/skills-for-fabric>

/plugin marketplace add microsoft/skills-for-fabric

/plugin install fabric-skills@fabric-collection

Or install a focused bundle

/plugin install powerbi-authoring@fabric-collection

CONNECT TO A SEMANTIC MODEL

1. Open Power BI Desktop
2. Connect to the SalesAnalytics semantic model with DirectLake
3. Save the file to the folder where you installed Copilot as a .pbib
4. Ask copilot to do things!



DEMO 4

LET'S CREATE A REPORT

LIVE WALKTHROUGH – FOLLOW ALONG IF YOU WANT OR WATCH & TAKE NOTES

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PART 3

FABRIC APP AND RAYFIN

What • setup • use cases
best practices • limits

Rayfin





Rayfin is a fully managed Backend-as-a-Service (BaaS) platform that helps teams build and ship applications faster without building or operating backend infrastructure.



DEMO 5

LET'S CREATE AN APP

LIVE WALKTHROUGH – FOLLOW ALONG IF YOU WANT OR WATCH & TAKE NOTES

CREATE A NEW APP ITEM

1. New Item → App → Give it a good name
 2. Copy the prompt and give it to your agent (easiest)
- Or
3. Copy the commands and paste in your terminal

BEST PRACTICES FOR AI-ASSISTED POWER BI

01



PROMPT WITH SCOPE

Define a specific, bounded task. Vague questions return vague answers.

02



VALIDATE ALL OUTPUT

Test, test, test! Confirm logic against the model before trusting.

03



SEPARATE RETRIEVAL FROM JUDGMENT

MCP retrieves. You decide. Never outsource the business meaning.

04



BUILD REPEATABLE WORKFLOWS

Define which prompts, tools and validation steps your team uses every time.

05



GOVERN COLLABORATION

Document AI-assisted outputs. Shared projects need shared accountability.

AI IS A LAYER, NOT A REPLACEMENT



The tool has changed. The need for judgment, domain knowledge, and business context hasn't.



AI accelerates understanding — but a confident-sounding wrong answer is still a wrong answer. Validate everything.



Governance matters more, not less. Faster output requires clearer ownership of what gets reviewed and approved.

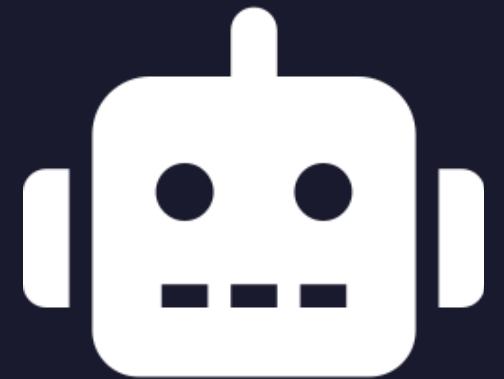


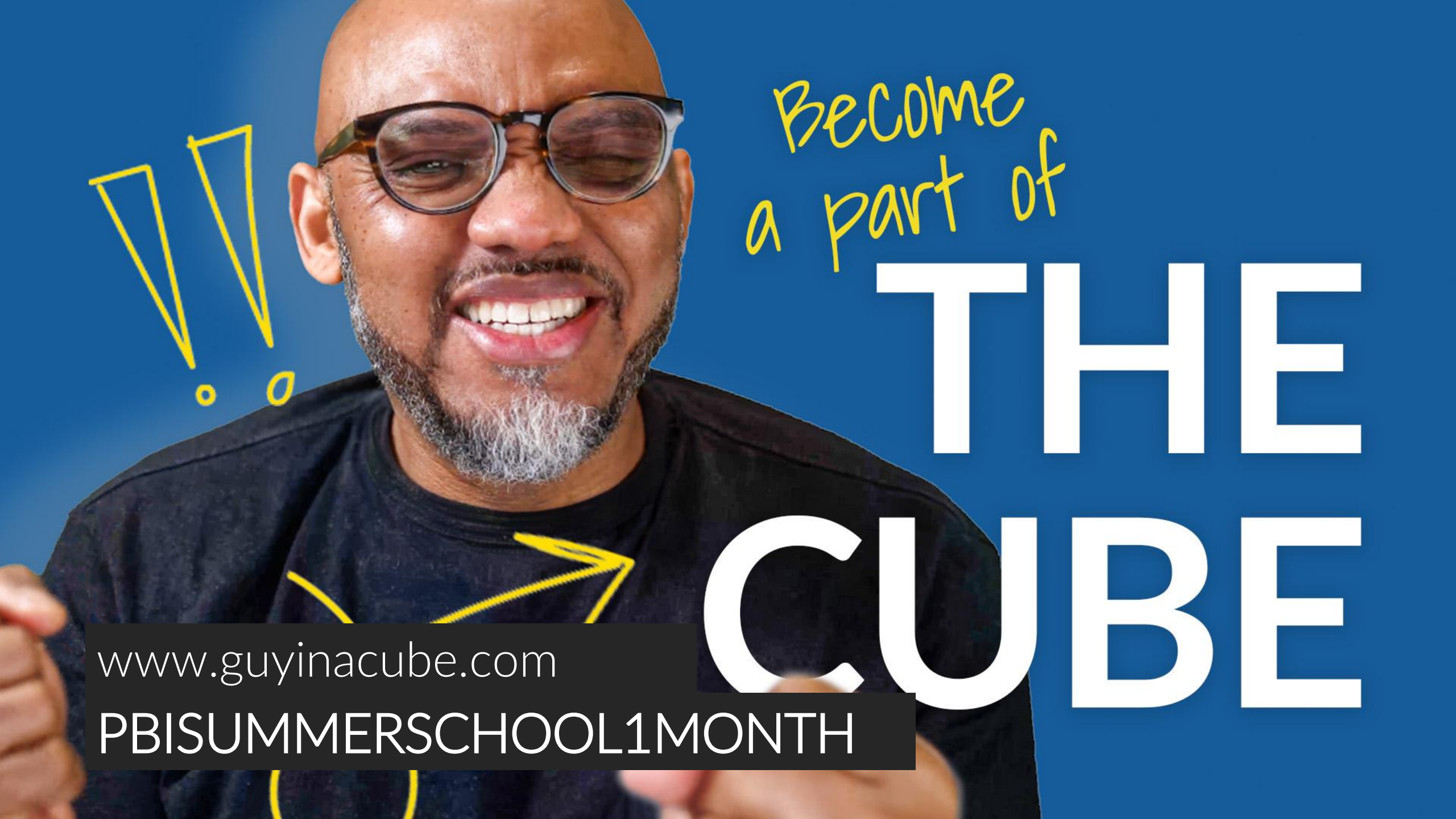
Start small. One good use case done well builds more trust than many half-baked workflows.

QUESTIONS?

THANKS FOR JOINING

- Copilot works best on well-structured, well-documented models
- MCP is governance + context + AI — not magic
- Retrieve with structure. Decide with judgment. Validate before you trust.





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